AQ5.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
The null space of a matrix $A_{4\times 3}$ A4×3 is

- ○ the subspace $W = \{x \in R^4 \mid Ax = 0\}$W = {x ∈ R4 | Ax = 0} of R^4R4.
- ○ the subspace $W = \{x \in R^3 \mid Ax = 0\}$W = {x ∈ R3 | Ax = 0} of R^3R3.
- ○ the subspace $W = \{x \in R^2 \mid Ax = 0\}$W = {x ∈ R2 | Ax = 0} of R^2R2.
- ○ the first column of the matrix $A_{4\times 3}$A4×3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
the subspace $W = \{x \in R^3 \mid Ax = 0\}$W = {x ∈ R3 | Ax = 0} of R^3R3.

1 point
Nullity of a matrix $A_{3\times 4}$A3×4 is

- ○ 3
- ○ 4
- ○ the number of independent variables in the system of linear equations $Ax = 0$Ax = 0, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ x = [x1x2x3x4].
- ○ the number of dependent variables in the system of linear equations $Ax = 0$Ax = 0, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$x = [x1x2x3x4].

No, the answer is incorrect.
Score: 0
Accepted Answers:
the number of independent variables in the system of linear equations $Ax = 0$Ax = 0, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

x = [x1x2x3x4] .

Consider the system of linear equations $Ax = 0$Ax = 0, where $x = [x_1\ x_2\ x_3]^T$ x = [x1x2x3]T and

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.A = [100000001] .

Use the above information to answer questions 3, 4 and 5.

1 point

Choose the correct statement from the following options.

- ○ x_1 x1, x_2 x2 and x_3 x3 all are dependent variables.
- ○ x_2 x2 is the only dependent variable.
- ○ x_1 x1 and x_3 x3 are the dependent variables and x_2 x2 is the only independent variable.
- ○ x_1 x1, x_2 x2 and x_3 x3 all are independent variables.

No, the answer is incorrect.
Score: 0
Accepted Answers:

x_1 x1 and x_3 x3 are the dependent variables and x_2 x2 is the only independent variable.

1 point

Which one of the following vector spaces represents the null space of AA?

- ○ $\{(t_1, 0, t_2) \mid t_1, t_2 \in R\}${(t1, 0, t2) | t1, t2 ∈ R}
- ○ $\{(0, t, 0) \mid t \in R\}${(0, t, 0) | t ∈ R}
- ○ $\{(t_1, t_2, 0) \mid t_1, t_2 \in R\}${(t1, t2, 0) | t1, t2 ∈ R}
- ○ $\{(0, 0, t) \mid t \in R\}${(0, 0, t) | t ∈ R}

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{(0, t, 0) \mid t \in R\}${(0, t, 0) | t ∈ R}

What will be the nullity of AA?

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 1

1 point

Find the nullity of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

1 point
Let S denote the set of solutions of the homogeneous system of linear equations $Ax = 0$. Which of the following statements is (are) true?

- [ ] If x_1 and x_2 are in S, then $x_1 - x_2$ is in S.
- [ ] If x is in S, then cx is in S, for any $c \in R$.
- [ ] If x_1 and x_2 are in S, then $\alpha x_1 + \beta x_2$ is in S, for any $\alpha, \beta \in R$.
- [ ] If x is in S, then $A^n x = 0$ for any $n \in N \setminus \{0\}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If x_1 and x_2 are in S, then $x_1 - x_2$ is in S.
If x is in S, then cx is in S, for any $c \in R$.
If x_1 and x_2 are in S, then $\alpha x_1 + \beta x_2$ is in S, for any $\alpha, \beta \in R$.
If x is in S, then $A^n x = 0$ for any $n \in N \setminus \{0\}$.

Find out the value of a for which the matrix $\begin{bmatrix} 1 & 2 \\ 3 & a \end{bmatrix}$ has nullity 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point

Level 2

1 point
Consider the coefficient matrix A of the following system of linear equations:

$$x_1 + x_2 + x_4 = 0$$
$$x_2 + x_3 = 0$$
$$x_1 - x_3 + x_4 = 0$$

Which one of the following vector spaces represents the null space of A?

○ $\{(t_1 + t_2, t_1, t_1, t_2) \mid t_1, t_2 \in R\}$

○ $\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$

○ $\{(t_1 + t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$

○ $\{(t_1 - t_2, -t_1, t_1, -t_2) \mid t_1, t_2 \in R\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$

Find the nullity of the matrix A, where $A = [a_{ij}]$ is of order 3×3 and $a_{i,j} = \min\{i, j\}$, $i, j = 1, 2, 3$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

1 point

Choose the correct set of options from the following.

- [ ] The nullity of a non-zero scalar matrix of order 3 must be 3.
- [ ] The nullity of a non-zero scalar matrix of order 3 must be 0.
- [ ] The nullity of a non-zero diagonal matrix of order 3 must be 3.
- [ ] The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The nullity of a non-zero scalar matrix of order 3 must be 0.
The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

Consider the system $Ax = b$, where $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$. The number of solutions of when $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

Check Answers